MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR MONOLITHIC AND MULTICHIP, SILICON
12 BIT DIGITAL-TO-ANALOG CONVERTERS

This amendment forms a part of MIL-M-38510/121A, dated 4 June 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

TABLE I, delete and substitute the following:

"

Gain error (TTL)	A_{E1}	All inputs = Logic "1" 3/ 7/ T_A = 25°C	T_A = 25°C	01,03, 04	-.20	+.20	%
				02	-.16	+.16	"
			T_A = 125°C	01	-0.25	+0.25	"
				02	-0.46	+0.46	"
				03	-0.50	+0.50	"
				04	-0.30	+0.30	"
			T_A = -55°C	01	-0.24	+0.24	"
				02	-0.40	+0.40	"
				03	-0.44	+0.44	"
				04	-0.28	+0.28	"

"

TABLE I, delete and substitute the following:

"

Logic "1" input current	I_{IH}	V_{CC} = 15 V, V_{IN} (logic "1") = 10.5 V, Each input measured separately V_{IN} (logic "1") = 5.5 V	01, 02, 03 04	-1 -300 145	+100 300 300	µA µA µA
Logic "0" input current	I_{IL}	V_{CC} = 15 V, V_{IN} (logic "0") = 0 V, Each input measured separately	01, 02, 03 04	-200 -100 5	+1 100 100	µA µA µA

"

PAGE 4

TABLE I, delete and substitute the following:

"

Bipolar offset error	B_{POE}	All inputs logic "0" 4/ 7/ T_A = 25°C		01,03, 04	-8	+8	LSB
				02	-6.5	+5.5	"
			T_A = 125°C	01	-10.0	+10.0	"
				02	-18.5	+18.5	"
				03	-20.0	+20.0	"
				04	-12.0	+12.0	"
			T_A = -55°C	01	- 9.6	+ 9.6	"
				02	-15.1	+15.1	"
				03	-17.6	+17.6	"
				04	-11.2	+11.2	"

"

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-01-14 15:36Z
Check the source to verify that this is the current version before use.

TABLE I, delete and substitute the following:

"

Bipolar zero error (TTL)	BZE	Input bits = 4,000 (octal), T_A = 25°C 7/	01	-6.5	6.5	LSB
			02	-6.5	6.5	"
			03,04	-6.0	6.0	"
		T_A = 125°C	01	- 8.5	+ 8.5	"
			02	-10.5	+10.5	"
			03,04	-10.0	+10.0	"
		T_A = -55°C	01	- 8.1	+ 8.1	"
			02	- 9.7	+ 9.7	"
			03,04	- 9.2	+ 9.2	"

"

PAGE 5

TABLE I, note 7, lines 2 and 3: Delete "4.5.8.1.3, 4.5.8.1.1, 4.5.8.1.2" and substitute "4.5.9.3, 4.5.9.1, 4.5.9.2" respectively.

PAGE 6

3.2.4, delete as substituted in amendment 1 and substitute the following:

"Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request."

TABLE II, delete and substitute the following:

"TABLE II. Electrical test requirements

MIL-STD-883 test requirement	Subgroups (see table III) 1/	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1, 2, 3, 4	1, 2, 3, 4
Group A test requirements (method 5005)	1,2,3,4,5,6, 7,8,12	1,2,3,4,5,5 7,9,12
Group C end-point and group B, class S, electrical parameters (method 5005)	1, 2, 3, 4, 5, 6, and table IV delta limits	1, 4, and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	---
Group D end-point electrical parameters (method 5005)	1, 2, 3, 4, 5, 6	1, 4

1/ PDA applies to subgroups 1 (see 4.2d)."

PAGES 35 THRU 39

FIGURE 6 - Continued: Delete and substitute new figure 6 - continued on pages 5 thru 9 of this amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

PAGE 45

TABLE III, device types 01 and 02, test number 44, limits columns: Delete "-0.20, 0.20" and "-0.16, 0.16" and substitute "-0.25, 0.25" and "-0.46, 0.46" respectively.

TABLE III, device types 01 and 02, test number 45, limits columns: Delete "-9.0, 8.0" and "-5.5, 6.5" and substitute "-10.0, 10.0" and "-18.5, 18.5" respectively.

TABLE III, device types 01 and 02, test number 46, limits columns: Delete "-6.5, 6.5" and "-6.5, 6.5" and substitute "-8.5, 8.5" and "-10.5, 10.5" respectively.

PAGE 47

TABLE III, device types 01 and 02, test number 87, limits columns: Delete "-0.20, 0.20" and "-0.16, 0.16" and substitute "-0.24, 0.24" and "-0.4, 0.4" respectively.

TABLE III, device types 01 and 02, test number 88, limits columns: Delete "-8.0, 8.0" and "-6.5, 6.5" and substitute "-9.6, 9.6" and "-16.1, 16.1" respectively.

TABLE III, device types 01 and 02, test number 89, limits columns: Delete "-6.5, 6.5" and "-6.5, 6.5" and substitute "-8.1, 8.1" and "-9.7, 9.7" respectively.

PAGE 50

TABLE III, device types 01 and 02, subgroup column: Delete "T_A = 55°C" and substitute "T_A = 25°C".

PAGE 57

* TABLE III, test numbers 5 through 16, for device type 03 only; Add "±100" in the limits columns. For device type 04, the limits remain the same as printed. Make this same change on page 63, test numbers 166 through 177 and on page 64, test numbers 199 through 210.

* TABLE III, test numbers 17 through 28 for device type 03 only: Add "±300" in the limits columns. For device type 04 the limits remain as printed. Make this same change on page 63, test numbers 178 through 189 and on page 64, test numbers 211 through 222.

PAGE 58

* TABLE III, test numbers 33 through 35, adapter pins column 4: Delete "+5 V" and substitute "+15 V". Make this same change on page 64, test numbers 194 through 196 and on page 65, test numbers 227 through 229.

* TABLE III, test number 35, adapter pin column 6: Delete "-10.5 V" and substitute "-16.5".

* TABLE III, test number 36, maximum limits column: Delete "10.0" and substitute "10.1".

TABLE III, device types 03 and 04, test number 38, limits columns: Delete "-0.20, 0.20" and "-0.20, 0.20" and substitute "-0.50, 0.50" and "-0.30, 0.30" respectively.

TABLE III, device types 03 and 04, test number 39, limits columns: Delete "-8, 8 and -8, 8" and substitute "-20, 20" and "-12, 12" respectively.

TABLE III, device types 03 and 04, test number 40, delete in its entirety.

TABLE III, device types 03 and 04, test number 41, limits columns: Delete "-6.0, 6.0" and substitute "-10, 10".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

PAGE 60

TABLE III, test number 81, limits columns: Delete "-0.20, 0.20" and substitute "-0.44, 0.44" for device type 03 and "-0.28, 0.28" for device type 04.

TABLE III, test number 82, limits columns: Delete "-8.0, 8.0" and substitute "-17.6, 17.6" for device type 03 and "-11.2, 11.2" for device type 04.

TABLE III, test number 83, delete in its entirety.

TABLE III, test number 84, limits columns: Delete "-6.0, 6.0" and substitute "-9.2, 9.2".

PAGE 67

4.2c and d, delete and substitute the following:

"c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the appliable device class."

4.4.1c, delete and substitute the following:

"c. A special subgroup shall be added to group A inspection for all classes and it shall consist of the conditions and limits as specified in table III, subgroup 12. LTPD shall be 15."

PAGE 68

4.4.3.6: Delete in its entirety.

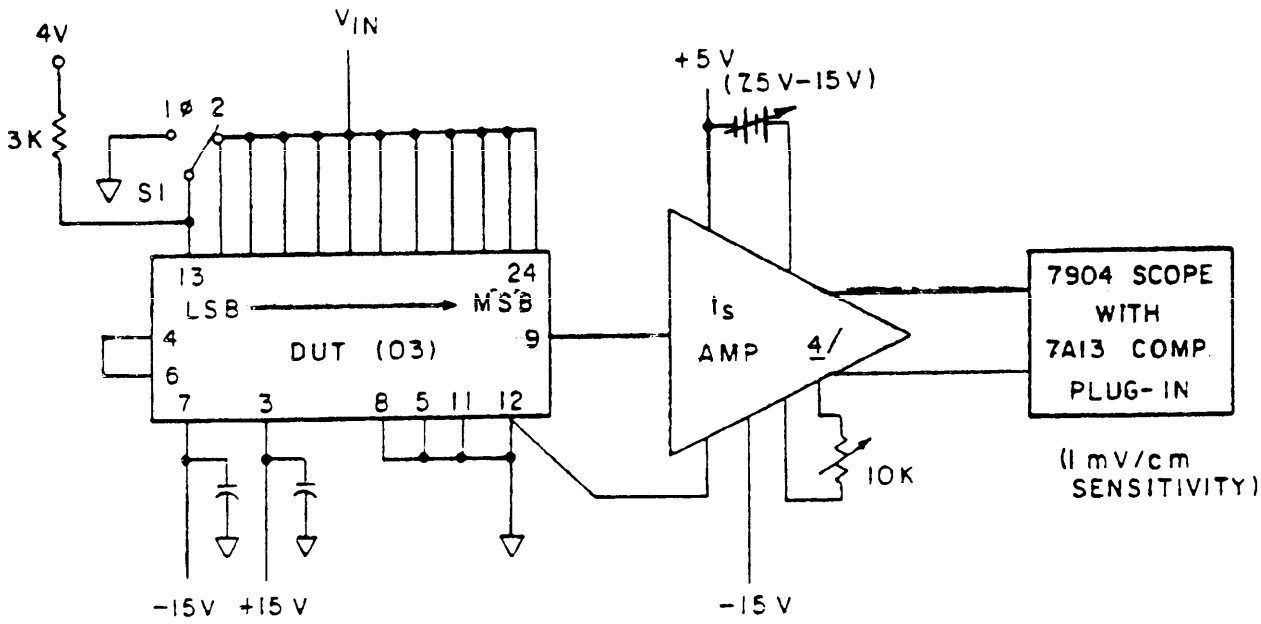
4.5.2, delete and substitute the following:

"Life test and burn-in cooldown procedure. When devices are measured at 25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to removal of the bias."

The margin of this amendment is marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Source: https://assist.dla.mil -- Downloaded: 20260-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 03



Device type 04

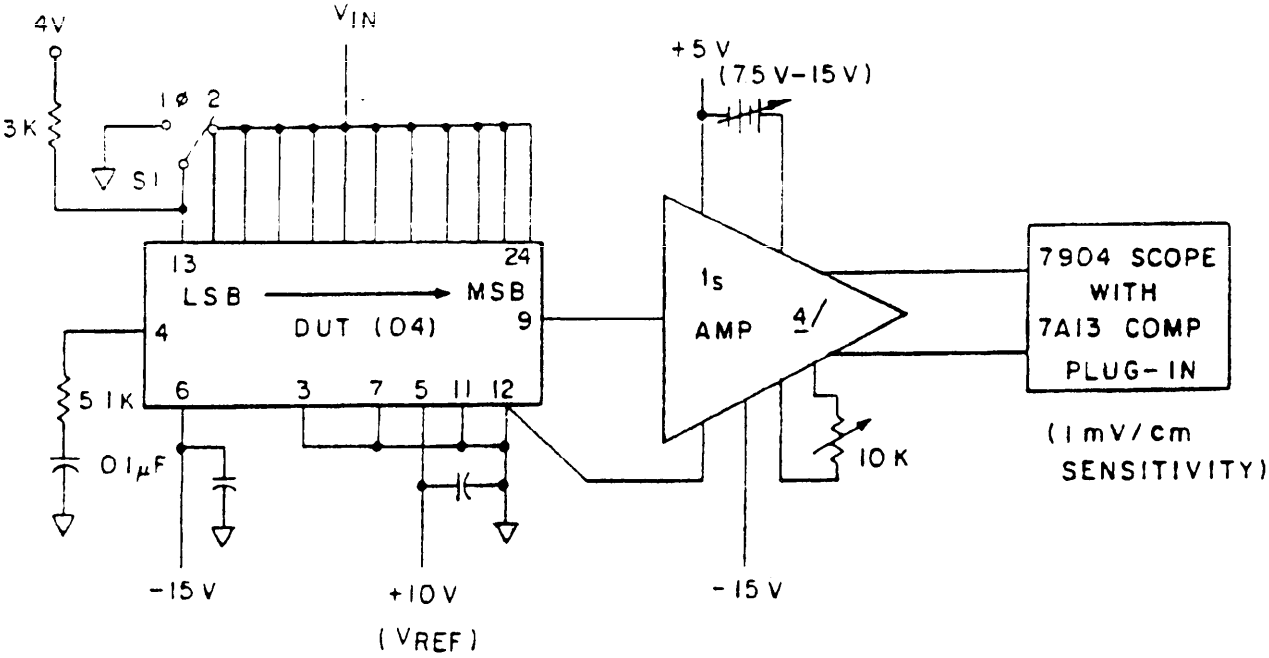
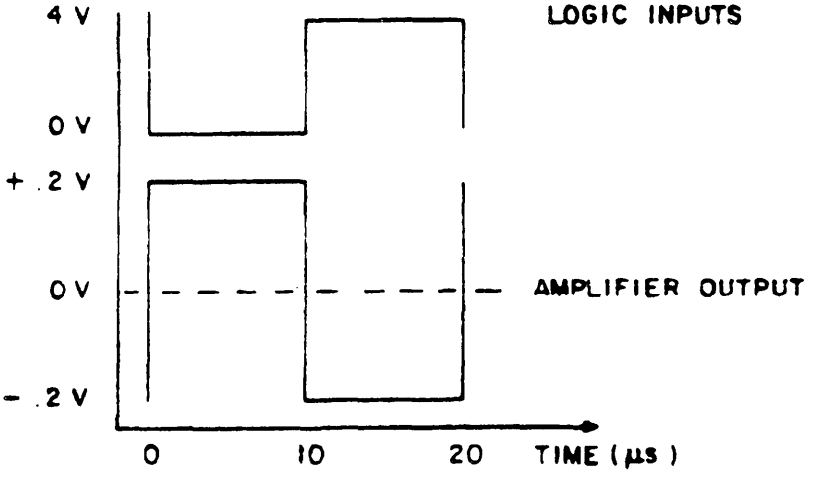


FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

NOTES:
1. Unless otherwise specified, all capacitors are 47 µF ceramic.
2. The bandwidth of the oscilloscope main frame must be at least 500 MHz.
3. The plug-in, given a 400 mV to 0 input pulse, must recover to 0.5 mV on the 1 mV/cm gain range within 100 ns.
4. See figure 7A for details of the settling time amplifier. For proper readings, the input FET's (Q1, Q2) and the output FET's (Q4, Q5) must be balanced by the following procedure:
   a. With the DUT removed from the socket, measure the gate voltage of Q1 with a DVM while adjusting the 2 kΩ trim potentiometer for a reading of 0 V.
   b. Connect the (+) and (-) output jacks to a differential comparator scope preamp. Next, zero the scope display with the scope switch on "GND". Short the Schottky diodes out and adjust the 5 kΩ potentiometer so that the differential output display is in the ground reference position, with the scope switch on "DC".
5. Calibrate the scope display by installing a DUT and applying a 4 V, 100 kHz square wave to the logic inputs. With the plug-in controls set to 200 mV/cm and 20 µs/cm, set the variable supply to 12.5 V and adjust the 10 kΩ pot until the amplifier output has its maximum amplitude. It should be a square wave approximately ±200 mV symmetrical about zero volts.

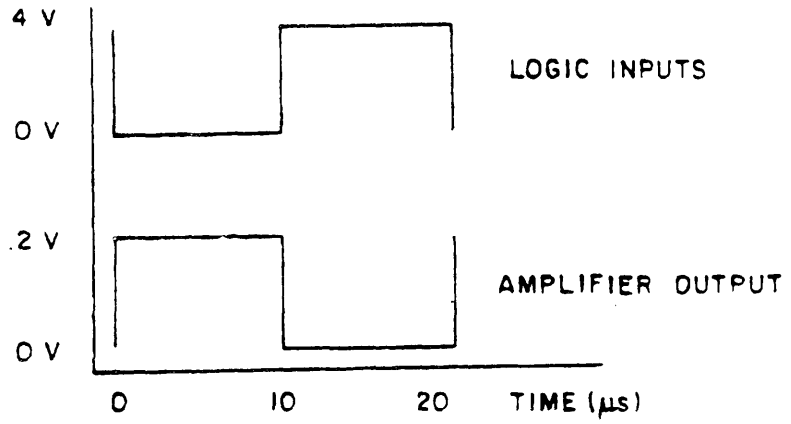


Measurement procedure:
1. Set the variable supply to 15 V and adjust the 10 kΩ pot such that the bottom of the square wave is at zero volts. Note that at 0 V the Schottky diodes are not clamping.

FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
Check the source to verify that this is the current version before use.

5. (Cont):



2. Increase the vertical plug-in gain to 1 mV/cm and the horizontal gain to 1 µs/cm. Toggle S1 to ground and observe the shift in the amplifier output. This is the value of 1 LSB and should be approximately 2.4 mV With S1 in position 2, observe the amplifier output level at 10 µs. It should be completely settled and unchanged from the 1 µs value. Any significant magnitude variations must be resolved before proceeding to measure DUT settling time. (If a significant change between 1 and 10 µs is noticed, it is probably due to a thermal unbalance in the amplifier fixture rather than a problem with the DUT. To check this, perform a t_{SHL} measurement between 1 and 10 µs. If the change is similar but of opposite polarity, it can be attributed to the fixture and ignored.)

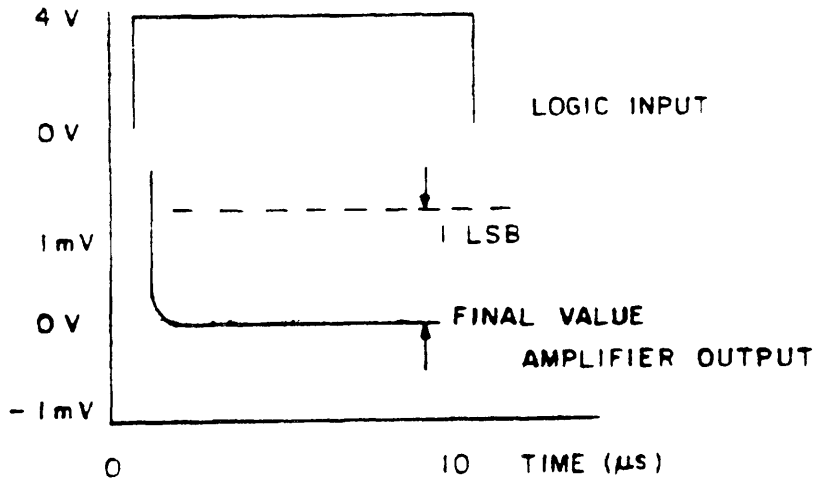
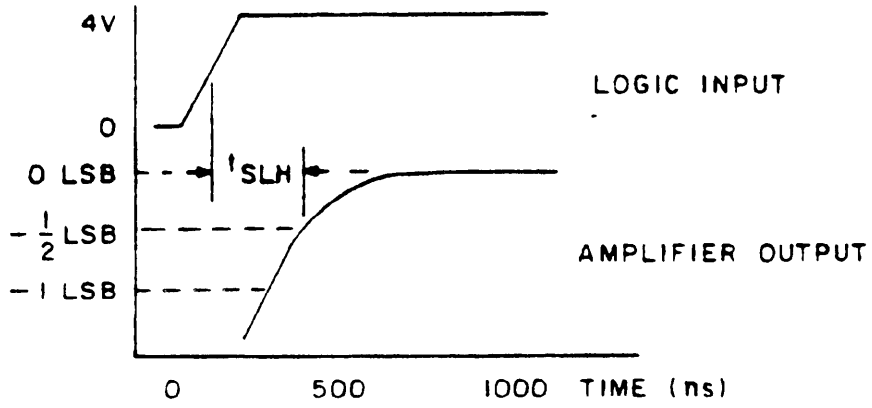


FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
Check the source to verify that this is the current version before use.

6. Measure the settling time by changing the horizontal plug-in to 100 ns/cm and observing when the amplifier output gets within 1/2 LSB of the 1.0 us final value.



7. Obtain t_{SHL} by adjusting the variable supply to 7.5 V and the 10 KΩ pot to zero the top of the amplifier output square wave and repeating the measurement, with the exception that S1 should be toggled to the Ø position to observe the value of a LSB.

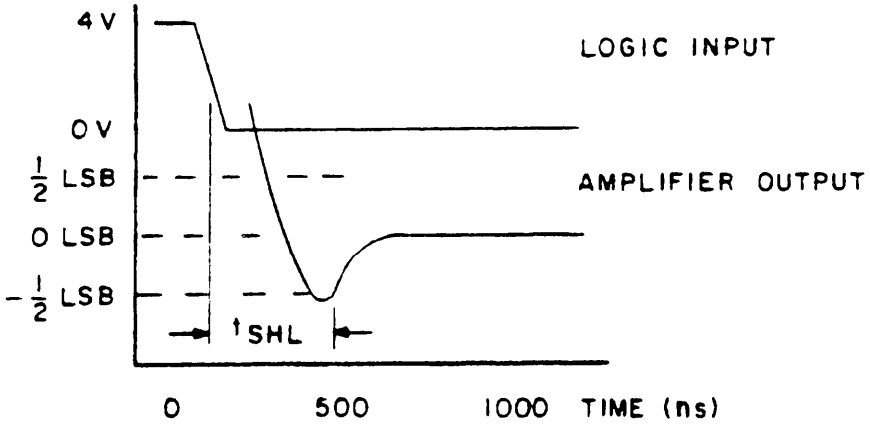
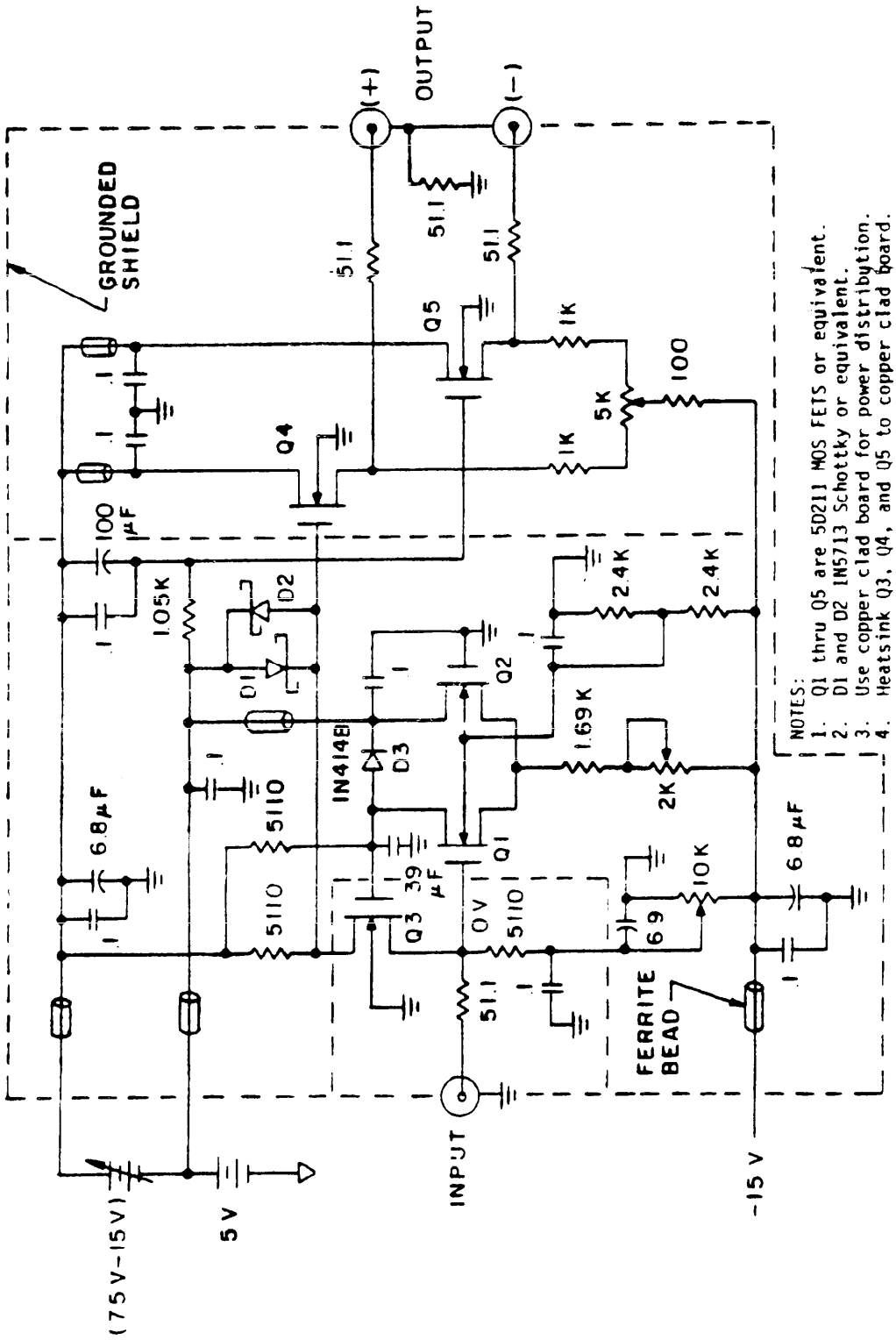


FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device types 03 and 04



NOTES:
1. Q1 thru Q5 are 5D211 MOS FETS or equivalent.
2. D1 and D2 IN5713 Schottky or equivalent.
3. Use copper clad board for power distribution.
4. Heatsink Q3, Q4, and Q5 to copper clad board. Attach a common to Q1 and Q2 electrically isolated from the PC board.

FIGURE 6. Test circuit for setting time. (amplifier)

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Civil agency coordinating activity:
NASA - NA

Preparing Acitivty:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0989)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.